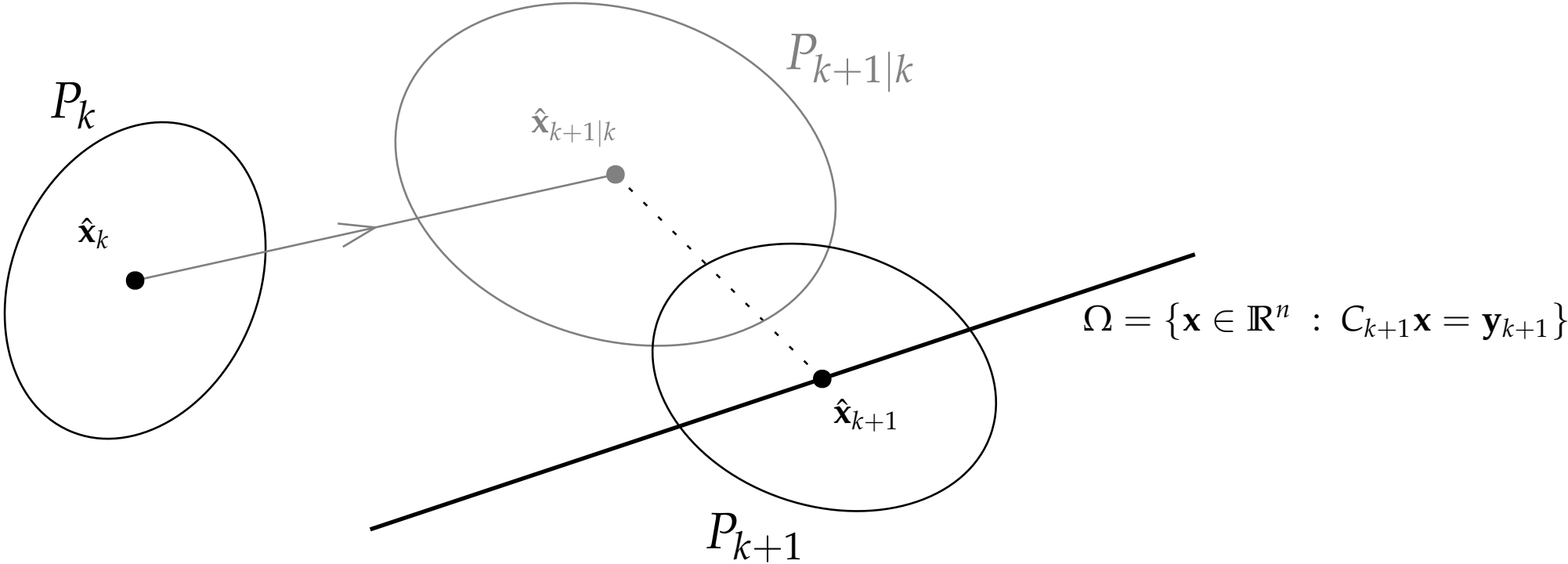
P_k
$\hat{\mathbf{x}}_k$
$P_{k+1|k}$
$\hat{\mathbf{x}}_{k+1|k}$
$\hat{\mathbf{x}}_{k+1}$
P_{k+1}
$\Omega = \{\mathbf{x} \in \mathbb{R}^n \ : \ C_{k+1}\mathbf{x} = \mathbf{y}_{k+1}\}$